Adjusted and frozen home packages Algorithm development

Our corrected workbook gives a clean package universe of **28 home packages** arranged as:

- **Home Economy 3kVA** → 1 version
- **Home Luxury 5kVA** → 6 versions
- **Home Deluxe 8kVA** → 4 versions
- **Ultra Power 10kVA** → 3 versions
- **Premium Power 12kVA** → 6 versions
- **Pro Power 16kVA** → 4 versions
- **Master Power 20kVA** → 4 versions

That is the structure the engine should now be built around.

The best foundation

The package engine should now be **package-first**, not inverter-first.

Meaning:

- do **not** build the logic around "5kVA family = this answer"
- build it around **actual sellable package records**
- then let PP, EP and user behavior route the client to the best three package records

That is the strongest long-term architecture.

1. The core design principle

The engine should work in this order:

Step 1

Customer selects appliances

Step 2

System calculates:

- **Effective PP**
- **Effective EP**

Step 3

System interprets:

- load management tolerance

- backup style
- smart-load eligibility
- phase requirement

Step 4

System scores all available packages

Step 5

System returns:

- **Budget**
- **Good Fit**
- **Excellent**

So the engine no longer asks:

“Which inverter family fits?”

It asks:

“Which three actual packages best fit this customer?”

That is the best solution.

2. Freeze the package master list

We now have one **master package table** with one row per package.

Use these permanent internal codes:

Home Economy

- HE-1

Home Luxury

- HL-1
- HL-2
- HL-3
- HL-4
- HL-5
- HL-6

Home Deluxe

- HD-1
- HD-2
- HD-3

- HD-4

Ultra Power

- UP-1
- UP-2
- UP-3

Premium Power

- PP-1
- PP-2
- PP-3
- PP-4
- PP-5
- PP-6

Pro Power

- PRO-1
- PRO-2
- PRO-3
- PRO-4

Master Power

- MP-1
- MP-2
- MP-3
- MP-4

These codes run the system.
Customers only see the **marketing names**.

The actual package structure will be as follows:

Below is the clean package structure from your finalized workbook.

Home Economy

Code	Inverter	Battery	Panels
HE-1	3kVA Must 24V	1 × 24V 100Ah Dyness	4 × 450W

Home Luxury

Code	Inverter	Battery	Panels
HL-1	6kVA Growatt	1 × 51V 100Ah Dyness	4 × 620W
HL-2	6kVA Growatt	1 × 51V 100Ah Dyness	6 × 620W
HL-3	6kVA Growatt	2 × 51V 100Ah Dyness	8 × 620W
HL-4	5kVA Sunsynk	1 × 51V 100Ah Dyness	6 × 620W
HL-5	5kVA Sunsynk	2 × 51V 100Ah Dyness	8 × 620W
HL-6	5kVA Sunsynk	1 × 51V 280Ah Dyness	10 × 620W

Home Deluxe

Code	Inverter	Battery	Panels
HD-1	8kVA Sunsynk	1 × 51V 200Ah Dyness	10 × 620W
HD-2	8kVA Sunsynk	1 × 51V 280Ah Dyness	12 × 620W
HD-3	8kVA Sunsynk	2 × 51V 200Ah Dyness	14 × 620W
HD-4	8kVA Sunsynk	2 × 51V 280Ah Dyness	16 × 620W

Ultra Power

Code	Inverter	Battery	Panels
UP-1	10kVA Sunsynk	1 × 51V 280Ah Dyness	14 × 620W
UP-2	10kVA Sunsynk	2 × 51V 200Ah Dyness	16 × 620W
UP-3	10kVA Sunsynk	2 × 51V 280Ah Dyness	21 × 620W

Premium Power

Code	Inverter	Battery	Panels
PP-1	12kVA Sunsynk 1P	1 × 51V 280Ah Dyness	14 × 620W
PP-2	12kVA Sunsynk 1P	2 × 51V 200Ah Dyness	18 × 620W
PP-3	12kVA Sunsynk 1P	2 × 51V 280Ah Dyness	24 × 620W
PP-4	12kVA Sunsynk 3P	1 × 51V 280Ah Dyness	14 × 620W
PP-5	12kVA Sunsynk 3P	2 × 51V 200Ah Dyness	18 × 620W
PP-6	12kVA Sunsynk 3P	2 × 51V 280Ah Dyness	24 × 620W

Pro Power

Code	Inverter	Battery	Panels
PRO-1	16kVA Sunsynk	2 × 51V 200Ah Dyness	18 × 620W
PRO-2	16kVA Sunsynk	2 × 51V 280Ah Dyness	21 × 620W
PRO-3	16kVA Sunsynk	4 × 51V 200Ah Dyness	27 × 620W
PRO-4	16kVA Sunsynk	4 × 51V 280Ah Dyness	30 × 620W

Master Power

Code	Inverter	Battery	Panels
MP-1	20kVA Sunsynk	2 × 51V 280Ah Dyness	24 × 620W
MP-2	20kVA Sunsynk	4 × 51V 200Ah Dyness	32 × 620W
MP-3	20kVA Sunsynk	4 × 51V 280Ah Dyness	40 × 620W
MP-4	20kVA Sunsynk	6 × 51V 280Ah Dyness	52 × 620W

4. What the developers must store for each package

Every package row must have these fields:

Field	Meaning
package_code	internal unique code
family_name	Home Economy / Home Luxury / etc
family_rank	1 to 7
variant_rank	1,2,3... inside family
inverter_size_kva	3, 5, 8, 10, 12, 16, 20
inverter_brand	Must / Growatt / Sunsynk
battery_total_kwh	actual total usable/nominal block used by your business
battery_type	24V100Ah / 51V100Ah / 51V200Ah / 51V280Ah
battery_count	quantity
panel_count	quantity
panel_wattage	450 / 620
array_kwp	panel_count × panel_wattage / 1000
phase	1P or 3P
smart_load_supported	Yes / No
net_metering_capable	Yes / No
material_cost	from sheet
labour_cost	from sheet
transport_cost	from sheet
total_price	from sheet
active	Yes / No

This is the **source of truth**.

5. Now redefine the engine around package capability

This is the crucial shift.

Instead of assigning PP/EP only to inverter sizes, assign **capability bands to each package**.

Each package gets:

Field	Meaning
pp_min	lowest effective PP this package is suitable for
pp_max	highest effective PP this package is suitable for
ep_min	lowest effective EP this package is suitable for
ep_max	highest effective EP this package is suitable for
recharge_class	Basic / Moderate / Balanced / Strong / Premium
comfort_class	Budget / Balanced / Premium
management_tolerance	High / Medium / Low

6. The routing logic

The engine should use:

Appliance table

Each appliance has:

- PP
- EP
- Concurrency
- Night Use
- Smart Load Eligible

Effective load formulas

- **Effective PP = PP × Concurrency**
- **Effective EP = EP × Night Use**

Then the engine scores every package

7. The package recommendation rule

For each client, calculate:

- effective_pp_total
- effective_ep_total
- smart_load_ratio
- user load-management preference
- phase requirement

Then:

Budget

Return the **lowest package that can still acceptably carry** the customer.

Good Fit

Return the **best balanced package** for performance, comfort, and market realism.

Excellent

Return the **next stronger package** above Good Fit.

This makes the labels **dynamic**.

That is exactly what we wanted:

- a package may be Budget for one client
- the same package may be Excellent for another

8. The scoring template recommended

Each package gets a fit score:

Package Score =

- **PP fit** → 40%
- **EP fit** → 35%
- **PV / recharge fit** → 15%
- **management fit** → 10%

So the algorithm is not random.
It is weighted.

9. The family anchor ranges

These are not the final answer — they are the **starting bands**.

Family	Effective PP anchor	Effective EP anchor
Home Economy	0.0 – 3.2	0.0 – 3.0
Home Luxury	2.8 – 7.8	3.0 – 13.0
Home Deluxe	6.5 – 11.8	8.5 – 21.0
Ultra Power	9.5 – 14.5	10.5 – 24.0
Premium Power	12.0 – 18.5	10.5 – 28.0
Pro Power	17.0 – 24.5	16.0 – 36.0
Master Power	23.0 – 32.0	22.0 – 48.0

Then each **variant** sits inside its family band.

10. Variant-level starter capability map

Use this as the first tuning base.

Home Economy

Code	PP Range	EP Range
HE-1	0.0 – 3.2	0.0 – 3.0

Home Luxury

Code	PP Range	EP Range
HL-1	2.8 – 4.8	3.0 – 5.5
HL-2	3.2 – 5.2	3.5 – 6.5
HL-3	4.0 – 6.2	6.0 – 9.5
HL-4	3.5 – 5.8	3.5 – 6.5
HL-5	4.5 – 6.8	6.5 – 10.5
HL-6	5.5 – 7.8	9.5 – 13.0

Home Deluxe

Code	PP Range	EP Range
HD-1	6.5 – 8.8	8.5 – 12.5
HD-2	7.0 – 9.5	10.5 – 14.5
HD-3	8.0 – 10.8	13.5 – 18.5
HD-4	9.0 – 11.8	15.5 – 21.0

Ultra Power

Code	PP Range	EP Range
UP-1	9.5 – 12.2	10.5 – 15.0
UP-2	10.5 – 13.2	14.0 – 19.0
UP-3	11.5 – 14.5	18.0 – 24.0

Premium Power

Code	PP Range	EP Range
PP-1	12.0 – 14.8	10.5 – 16.0
PP-2	13.0 – 16.2	15.0 – 20.5
PP-3	14.0 – 17.5	20.0 – 28.0
PP-4	12.0 – 14.8	10.5 – 16.0
PP-5	13.0 – 16.2	15.0 – 20.5
PP-6	14.0 – 17.5	20.0 – 28.0

Pro Power

Code	PP Range	EP Range
PRO-1	17.0 – 20.5	16.0 – 22.0
PRO-2	18.0 – 21.5	20.0 – 26.0
PRO-3	20.0 – 23.5	24.0 – 32.0
PRO-4	21.0 – 24.5	28.0 – 36.0

Master Power

Code	PP Range	EP Range
MP-1	23.0 – 26.5	22.0 – 30.0
MP-2	25.0 – 29.0	28.0 – 38.0
MP-3	27.0 – 31.0	36.0 – 46.0
MP-4	29.0 – 32.0	44.0 – 55.0

These are **starter capability bands**, not sacred doctrine.
They give your developers a solid build target, and we tune them with real cases.

11. Very important override rules

These must be coded.

Rule A — Smart-load benefit

If a package has **smart_load_supported = Yes**, it may carry slightly more PP in Budget mode.

Rule B — Pump penalty

If the client has:

- borehole
- booster pump
- pressure pump

then Budget recommendations should not be too aggressive unless the user indicates they are happy to manage those loads.

Rule C — Heavy burst loads

Kettle, iron, toaster, microwave, hair dryer, vacuum cleaner:

- increase PP strongly
- but should not inflate EP too much

Rule D — No downgrade below acceptable

If no package fits exactly, the engine should go **up**, not down.

Table A : Master Appliance Database

Appliance	PP	EP	Concurrency	NightUse	SmartLoadEligible	Notes
LED Light (single)	0.2	0.5	0.8	1.0	No	low power lighting
LED lights group (5–10)	1	3	0.8	1.0	No	typical house lighting
TV (32–55")	1	2	0.6	0.8	No	evening entertainment
Home theatre	1	1.5	0.4	0.7	No	short use
WiFi router	0.3	2	1.0	1.0	No	runs continuously
Laptop	0.5	1	0.5	0.5	No	intermittent
Desktop computer	1	2	0.6	0.6	No	office workstation
Phone chargers	0.2	1	0.7	0.7	No	multiple devices
Refrigerator	2	3	0.6	0.6	No	compressor cycling
Deep freezer	2	3	0.5	0.6	No	similar to fridge
Microwave	4	1	0.3	0.2	No	short bursts
Electric kettle	5	1	0.2	0.1	No	very high burst
Toaster	3	1	0.2	0.1	No	short cooking
Electric frying pan	4	2	0.3	0.2	No	cooking appliance
Blender	2	0.5	0.2	0.1	No	short bursts
Coffee machine	2	1	0.3	0.2	No	short runtime
Washing machine	3	2	0.4	0.2	Yes	intermittent cycles
Dishwasher	3	2	0.4	0.2	Yes	intermittent
Iron	3	2	0.3	0.2	No	periodic use
Hair dryer	4	1	0.2	0.1	No	short bursts
Water pump (booster)	3	1.5	0.5	0.3	Yes	intermittent pump
Borehole pump	6	2	0.4	0.1	Yes	startup surge
Pool pump	3	2	0.3	0.1	Yes	longer runtime
Air conditioner (9000 BTU)	6	4	0.6	0.5	Yes	sustained cooling
Air conditioner (18000 BTU)	8	5	0.6	0.5	Yes	heavy cooling

Appliance	PP	EP	Concurrency	NightUse	SmartLoadEligible	Notes
Ceiling fan	0.5	1	0.7	0.7	No	continuous airflow
Pedestal fan	0.5	1	0.7	0.7	No	portable fan
Electric heater	7	6	0.5	0.8	No	heavy winter load
Electric stove plate	8	5	0.4	0.2	No	cooking
Electric oven	10	6	0.3	0.2	No	heavy cooking
Electric geyser	10	7	0.4	0.2	Yes	water heating
Solar geyser pump/controller	0.5	0.5	0.8	0.5	No	small circulation
CCTV system	0.5	2	1.0	1.0	No	24/7 security
Alarm system	0.3	1.5	1.0	1.0	No	always on
Gate motor	4	0.5	0.1	0.1	No	short bursts
Garage door motor	4	0.5	0.1	0.1	No	short bursts
Electric fence energizer	0.5	1	1.0	1.0	No	continuous
Printer	1	0.5	0.3	0.2	No	intermittent
Photocopier	3	1	0.2	0.1	No	office burst load
Server / network rack	2	4	1.0	1.0	No	continuous IT load
Shop POS system	1	2	1.0	0.8	No	retail
Borehole filtration system	2	2	0.4	0.2	Yes	water treatment
Water purifier	0.5	1	0.3	0.2	No	intermittent
Under-counter fridge	1	2	0.7	0.6	No	small refrigeration
Ice maker	2	2	0.5	0.4	No	intermittent
Tumble dryer	7	5	0.3	0.1	Yes	heavy cycle
Gaming console	1	2	0.6	0.7	No	evening entertainment
Projector	1	2	0.4	0.6	No	occasional use
Outdoor lighting	1	3	0.6	1.0	No	night lighting
Garden irrigation controller	0.5	1	0.4	0.2	Yes	irrigation system
Workshop tools (small)	2	1	0.3	0.1	No	intermittent
Angle grinder	4	1	0.2	0.1	No	short bursts
Welder (small inverter)	10	2	0.1	0.1	No	rarely used

SMART LOAD LOGIC MODULE (VERSION 1)

1. Purpose of this module

This module explains how the platform should handle **smart-load-capable appliances** when calculating solar package recommendations.

The goal is to make the recommendation engine:

- more realistic
- more commercially competitive
- more aligned with actual inverter capabilities
- easier for customers to understand

This is especially important when comparing packages using:

- **Sunsynk inverters**, which support smart load management
- **Growatt inverters**, which do not provide the same automatic smart-load control

---

2. Core concepts

The system must distinguish between two different ideas:

A. Smart Load Eligible

This belongs to the **appliance**.

It means the appliance is a type of load that can reasonably be:

- shifted to daytime
- restricted when battery is low
- switched off manually
- connected to an AUX / smart port
- deprioritized during backup operation

Examples:

- borehole pump
- pressure pump
- booster pump
- washing machine
- pool pump

- geyser
- filtration system
- some air conditioners

These are **not always essential 24/7 loads**.

B. Smart Load Supported

This belongs to the **package / inverter**.

It means the package includes an inverter that can actually support smart-load behavior automatically.

Examples:

- **Sunsynk package** → Smart Load Supported = Yes
- **Growatt package** → Smart Load Supported = No

This distinction is critical.

A load may be **smart-load eligible**, but if the package inverter does not support smart-load control, the system must not assume the same level of automation.

---

3. Why this matters

Without this logic, the calculator will either:

Oversize systems

If all loads are treated as always fully active

or

Undersize systems dangerously

If manageable loads are discounted too aggressively regardless of inverter type

The smart-load layer allows the engine to recommend:

- **lower-cost workable systems** where appropriate
- **stronger systems** where convenience is needed
- **Sunsynk packages** where automatic load control adds real value
- **Growatt packages** where manual discipline is acceptable

---

4. User questions required by the engine

Before the package engine calculates final recommendations, the system must ask the client a few practical questions.

These questions must be simple and non-technical.

Question 1 — Main priority

What matters most to you?

- Lowest workable cost
- Best balance of cost and performance
- Maximum comfort and performance

This determines the recommendation lane:

- Budget
- Good Fit
- Excellent

Question 2 — Load management willingness

Are you comfortable switching off some heavy appliances when battery is low?

- Yes
- No

This tells the system whether manual load management is acceptable.

Question 3 — Use style

How do you want the system to work?

- Mainly for backup
- Backup plus daytime solar savings
- As much solar independence as possible

This influences how conservative the system should be.

Question 4 — Heavy controllable loads

This question appears only if the client selected smart-load-eligible appliances.

Some of your selected appliances can be controlled or limited when battery is low. Would you like us to consider that in the recommendation?

- Yes
- No

This helps the system decide how much smart-load discount to apply.

---

5. Which appliances are smart-load eligible

This must be stored in Table A.

Typical smart-load-eligible appliances include:

- Borehole pump
- Pressure pump
- Booster pump
- Pool pump
- Washing machine
- Dishwasher
- Geyser
- Filtration system
- Garden irrigation controller
- Tumble dryer
- Some air conditioners

These appliances are considered manageable because they do not always need unrestricted battery priority.

---

6. Sequence of calculation

The system must follow this order exactly.

Step 1 — Collect appliance selections

User selects appliances.

Step 2 — Read appliance values from Table A

For each selected appliance, retrieve:

- PP
- EP
- Concurrency
- NightUse

- SmartLoadEligible

Step 3 — Calculate base effective values

For each appliance:

```
Base_Effective_PP = PP × Concurrency
Base_Effective_EP = EP × NightUse
```

At this point, no smart-load discount has yet been applied.

Step 4 — Determine recommendation mode

Based on the user's answer, set the recommendation lane:

- Budget
- Good Fit
- Excellent

Step 5 — Evaluate each candidate package

The engine must test packages one by one.

For each package, retrieve:

- package_code
- inverter_brand
- smart_load_supported
- pp_min / pp_max
- ep_min / ep_max
- price
- family_name

Step 6 — Apply smart-load logic per appliance per package

For each selected appliance:

If appliance is NOT smart-load eligible:

Use base values only.

```
Adjusted_PP = Base_Effective_PP
Adjusted_EP = Base_Effective_EP
```

If appliance IS smart-load eligible:

Then the engine checks:

1. Is the package Smart Load Supported?
2. Does the client accept manual load management?

Then apply the correct modifier.

Step 7 — Sum adjusted totals

After all selected appliances are processed:

```
Total_Adjusted_PP = sum of Adjusted_PP
Total_Adjusted_EP = sum of Adjusted_EP
```

Step 8 — Compare against package capability

Compare totals to Table C:

- PP range
- EP range
- management tolerance
- recharge class

Step 9 — Score packages

Rank packages based on fit.

Step 10 — Return final 3 recommendations

Return:

- Budget
- Good Fit
- Excellent

---

7. Smart-load modifier logic

This is the heart of the smart-load calculation.

Case 1 — Appliance is smart-load eligible AND package supports smart load

Example:

- Borehole pump
- Sunsynk package

Apply full smart-load advantage.

Budget mode

```
Adjusted_PP = Base_Effective_PP × 0.8
Adjusted_EP = Base_Effective_EP × 0.6
```

Good Fit mode

```
Adjusted_PP = Base_Effective_PP × 0.9
Adjusted_EP = Base_Effective_EP × 0.8
```

Excellent mode

```
Adjusted_PP = Base_Effective_PP × 1.0
Adjusted_EP = Base_Effective_EP × 1.0
```

Case 2 — Appliance is smart-load eligible, package does NOT support smart load, but client accepts manual management

Example:

- Borehole pump
- Growatt package
- Client is willing to switch it off manually

Apply a weaker discount.

Budget mode

```
Adjusted_PP = Base_Effective_PP × 0.9
Adjusted_EP = Base_Effective_EP × 0.75
```

Good Fit mode

```
Adjusted_PP = Base_Effective_PP × 0.95
Adjusted_EP = Base_Effective_EP × 0.9
```

Excellent mode

```
Adjusted_PP = Base_Effective_PP × 1.0
Adjusted_EP = Base_Effective_EP × 1.0
```

Case 3 — Appliance is smart-load eligible, package does NOT support smart load, and client does NOT accept manual management

Example:

- Borehole pump
- Growatt package
- Client wants convenience

Apply no discount.

```
Adjusted_PP = Base_Effective_PP
Adjusted_EP = Base_Effective_EP
```

---

8. Why EP usually gets more discount than PP

This is intentional.

A smart-load appliance may still matter for inverter sizing sometimes, but may not need to burden the battery all night.

Example:

- a borehole pump still affects inverter power
- but if it is not allowed to run at night, its battery burden should be much lower

So:

- PP discount = moderate
- EP discount = stronger

That gives the engine more realistic behavior.

---

9. How Sunsynk and Growatt are differentiated

The platform must clearly distinguish the two.

Sunsynk packages

Strengths:

- smart-load support
- AUX / smart-port behavior
- better managed-load handling
- better for pumps and heavy controllable loads
- stronger fit for budget systems with tricky appliances

Growatt packages

Strengths:

- lower entry price
- good budget systems where the user is disciplined
- suitable when client accepts manual management

Limitations:

- no equivalent automatic smart-load logic
- less suitable for complex pump-heavy or management-sensitive setups

This means the engine may recommend:

- **Growatt** as the lower-cost route
- **Sunsynk** as the better-controlled and more refined route

even when both are 5kVA.

---

10. Package recommendation philosophy

The system must not “prefer Sunsynk no matter what.”

It must make a fair recommendation based on:

- client budget
- client management willingness
- selected appliances
- package capability
- smart-load support

Example outcome

For one client:

- Budget → Growatt 5kVA
- Good Fit → Sunsynk 5kVA
- Excellent → Sunsynk 8kVA

For another client:

- Budget → Sunsynk 5kVA
- Good Fit → Sunsynk 8kVA
- Excellent → Sunsynk 10kVA

That is correct.

---

11. Developer rules summary

Rule 1

Smart-load modifiers are applied **per appliance**, not to the total formula.

Rule 2

Only appliances with `SmartLoadEligible = Yes` may receive smart-load discounts.

Rule 3

A full smart-load discount is only allowed if:

- package supports smart load
  or
- the user accepts manual management (then use reduced discount)

Rule 4

Growatt packages must not receive the same smart-load advantage as Sunsynk packages.

Rule 5

Excellent recommendations should generally use little or no smart-load discount.

Rule 6

If there is uncertainty, the engine should favor the safer package.

---

12. Minimal data requirements for developers

In Table A

Need:

- appliance_name
- pp
- ep
- concurrency
- night_use
- smart_load_eligible

In Table C

Need:

- package_code
- inverter_brand
- smart_load_supported
- pp_min
- pp_max
- ep_min
- ep_max
- management_tolerance

In recommendation logic

Need:

- client priority
- client willingness to manage loads

---

13. Recommended customer-facing wording

The website should explain this simply.

Example:

Some heavy appliances such as pumps, geysers, and washing machines can be controlled or reduced when battery is low.
If you are comfortable with that, the system can recommend a more affordable solar package.

This helps the client understand why one package may be cheaper than another.

---

14. Final implementation note

This smart-load layer should be treated as a **controlled recommendation adjustment**, not as an engineering override.

It improves:

- competitiveness
- realism
- customer fit

without making the system reckless.

TABLE C — PACKAGE CAPABILITY MATRIX

Purpose

Table C defines **what each package is capable of handling**.

It acts as the **bridge between appliance demand (PP & EP)** and the **actual solar packages offered on the website**.

Each package has a defined operating range for:

- **PP (Power Points)** → determines inverter sizing
- **EP (Energy Points)** → determines battery sizing
- **PV capacity** → determines solar charging performance

These ranges allow the algorithm to determine which packages can realistically support a given appliance set.

Package Families

The platform uses the following package families:

Package Family	Inverter Size
Home Economy	3kVA
Home Luxury	5kVA
Home Deluxe	8kVA
Ultra Power	10kVA
Premium Power	12kVA
Pro Power	16kVA
Master Power	20kVA

These correspond with the naming convention already used on the Taqon Electrico website to maintain **SEO continuity**.

Table C Structure

Each package record must include the following fields.

Field	Description
package_code	Unique identifier
package_name	Marketing name
inverter_size	kVA rating
inverter_brand	Sunsynk / Growatt
smart_load_supported	Yes / No
panel_count	Number of solar panels
battery_capacity	kWh
pp_min	Minimum PP range
pp_max	Maximum PP range
ep_min	Minimum EP range
ep_max	Maximum EP range
recharge_class	Solar recharge capability
management_tolerance	How much load management is assumed

Example Table C Entries

Code	Package Name	Inverter	Brand	Smart Load	Panels	Battery	PP Range	EP Range
HE-1	Home Economy	3kVA	Must	No	4	2.4kWh	0–3	0–3
HL-1	Home Luxury	5kVA	Growatt	No	4	5kWh	3–5	3–6
HL-2	Home Luxury	5kVA	Growatt	No	6	5kWh	3–5.5	3–7
HL-3	Home Luxury	5kVA	Growatt	No	8	10kWh	4–6	6–10
HL-4	Home Luxury	5kVA	Sunsynk	Yes	6	5kWh	3–6	3–7
HL-5	Home Luxury	5kVA	Sunsynk	Yes	8	10kWh	4–7	6–11
HL-6	Home Luxury	5kVA	Sunsynk	Yes	10	10kWh	5–8	8–13

Key Concept

The engine **does not directly recommend an inverter size first**.

Instead it:

1 Calculates **PP and EP demand**
2 Tests which packages fall within capability ranges

3 Scores packages
4 Returns best recommendations

TABLE D — PACKAGE RECOMMENDATION ENGINE

Table D defines **how the algorithm selects packages**.

It converts appliance selections into three recommended packages:

- **Budget**
- **Good Fit**
- **Excellent**

Step 1 — Appliance Input

The user selects appliances.

Example:

Appliance	Quantity
Lights groups	2
TV	2
Fridge	1
Borehole pump	1
WiFi	1

Step 2 — Read Appliance Data

From Table A retrieve:

- PP
- EP
- Concurrency
- NightUse
- SmartLoadEligible

Step 3 — Calculate Base Load

For each appliance:

```
Base_PP = PP × Concurrency
Base_EP = EP × NightUse
```

Step 4 — Apply Smart Load Logic

If appliance is smart-load eligible, apply modifiers depending on:

- inverter type
- user willingness to manage loads
- recommendation lane

Examples:

Sunsynk packages

Budget mode

```
PP × 0.8
EP × 0.6
```

Good Fit

```
PP × 0.9
EP × 0.8
```

Growatt packages

Budget mode

```
PP × 0.9
EP × 0.75
```

Good Fit

```
PP × 0.95
EP × 0.9
```

Step 5 — Sum Total Demand

```
Total_PP = Sum of Adjusted_PP
Total_EP = Sum of Adjusted_EP
```

These totals represent the **household demand profile**.

Step 6 — Test Against Package Capability

Each package in Table C is tested.

A package qualifies if:

```
Total_PP within package PP range
AND
Total_EP within package EP range
```

Step 7 — Package Scoring

Qualified packages are scored based on four criteria.

Factor	Weight
PP fit	40%
EP fit	35%
PV recharge capacity	15%
Smart-load compatibility	10%

Step 8 — Select Top Packages

From the ranked list:

Recommendation	Selection Rule
Budget	Lowest cost workable package
Good Fit	Highest scoring package
Excellent	Next stronger package above Good Fit

Example Output

```
Recommended Packages
```

Budget

5kVA Growatt – Home Luxury

Good Fit

5kVA Sunsynk – Home Luxury

Excellent

8kVA Sunsynk – Home Deluxe

Why This Structure Works

This method ensures:

✓ realistic inverter sizing
✓ realistic battery sizing
✓ smart load logic is respected
✓ Sunsynk advantages are recognized
✓ Growatt still remains competitive
✓ packages remain standardized

Developer Implementation Notes

The recommendation engine should be structured in four modules:

Module 1

Appliance scoring engine

Module 2

Smart load adjustment engine

Module 3

Package capability testing

Module 4

Package scoring and recommendation

Result

The platform will behave like an **experienced solar engineer**, automatically:

- interpreting appliance usage
- considering inverter capabilities
- recommending realistic systems
- generating automated quotations

Where the array still matters in the algorithm

Please note, the solar array is not dynamically designed by the calculator in Version 1.
The calculator determines the most suitable package, and the matched package already contains a predefined solar array configuration.

However, even though the array is tied to the package, it should still influence **package scoring**.

Meaning:

If two packages both match PP and EP reasonably, the engine can prefer the one whose solar array gives better recharge suitability for the recommendation lane.

Example:

- one package has 8 panels
- another has 10 panels
- both are possible
- the 10-panel option may score higher as **Good Fit** or **Excellent**

So the array is **not calculated independently**, but it still affects ranking.